

Course Number	0418-415	Course Title	Information Theory and Cryptography
Credit Hours	3 (3:0:3)	Course Coordinator	Dr. Ludovit Niepel
Compulsory	No	Prerequisite	0418-211

Textbook

<i>Title:</i>	<i>Elements of Information Theory</i>
<i>Author:</i>	<i>Th. M. Cover, J. A. Thomas</i>
<i>Publisher:</i>	<i>John Wiley & Sons, Latest Edition.</i>

Major Topics Covered in the Course

	Topics	Hours
1	Introduction: Security threats in the digital world, security services and mechanisms.	2
2	Elements of probability theory and concepts of information theory: Review basic laws of probability theory, conditional probabilities; independence; self-information; measure of information (entropy); joint and conditional entropies; properties of information measures; applications.	8
3	Communication systems and information channels: Memory-less source; binary sources; memory-less channels; channel capacity; losses, noise, redundancy and transmission efficiency; special channels: ideal, noiseless, lossless, uniform, binary symmetric, binary erasure, general binary channels; channels in cascade; data processing theorem; channels in parallel; extension of sources and channels; Muroga method.	10
4	Coding: Importance of coding; encoders and decoders; types of codes; encoding efficiency; redundancy of the code; special codes and their construction; Shannon-Fano and Huffman encoding procedures; error detection and error correction codes; Hamming codes.	8
5	Encryption methods for information protection and system security: Classical ciphers, private and public key, cryptography, authentication codes, ideal secrecy systems, practical security methods, data encryption standard, message authentication and signature schemes; application of cryptographic methods to operating systems, computer networks, and databases to achieve system security.	11
6	Exams.	3

	Total.	42
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Grading

20%	Midterm 1
20%	Midterm 2
20%	Hmw/quiz/project
40%	Final

Course Outcomes

Students should be able to:

1. Understand information measures of communication systems, different codes and encryption methods.
2. Implement several applications of information theory and crypto-graphic methods to computer networks and operating systems.
3. Distinguish between different information channels and to compute their capacities and the transmission efficiency.
4. Compare different codes, their practical use and the error detection and correction methods.
5. Identify and evaluate the professional, legal, social, security and ethical aspects of information, work in groups on a project

Relationship between Course Outcomes and Student Outcomes

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Course Outcome	Student Outcomes					
	1	2	3	4	5	6
1	H					
2	L					
3	L					
4	H					
5				M	M	